

Project Proposal: Modeling and Simulation

1. Project Title

Development of a Modeling and Simulation Framework for [System/Process Name]

2. Introduction

Modeling and simulation (M&S) are essential tools used to represent real-world systems in a virtual environment. These tools enable analysis, prediction, and optimization of system performance without costly or risky real-world experimentation. This project aims to design and implement a simulation model to better understand and improve a selected system.

3. Problem Statement

Complex systems are often difficult to evaluate using traditional methods. There is a need to apply modeling and simulation techniques to analyze system behavior, detect inefficiencies, predict performance under varying conditions, and support data-driven decision-making.

4. Objectives

- Develop a conceptual and computational model of the system
- Simulate different operational scenarios
- Analyze system performance using key indicators
- Provide recommendations for improvement
- Validate the model outputs

5. Scope of the Study

Focus on key processes and interactions within the system using simulation tools to evaluate performance. Limitations may include assumed or limited data and inability to capture all real-world uncertainties.

6. Methodology

Data Collection, Model Development, Simulation, Validation, and Analysis stages will be followed to complete the project.

7. Expected Outputs

A working simulation model, graphs and visualizations, analytical report, and improvement recommendations.

8. Tools and Technologies

Python, MATLAB, R, AnyLogic, Arena, Simulink, and data visualization tools.

9. Timeline

Planning & Data Collection (1–2 weeks), Model Development (2–3 weeks), Simulation & Testing (2 weeks), Analysis & Reporting (1–2 weeks).

10. Project Guidelines

Each group must consist of a maximum of five (5) members only. All members are expected to contribute actively. Each group must select a unique system or case study. Proper documentation and referencing are required. Outputs must include both the model and a written report. Presentations may be required.

11. Conclusion

This project highlights the importance of modeling and simulation in analyzing complex systems and improving decision-making.